

MUHAMMAD HAMZA RAFI

Web Developer

[LinkedIn](#) • [GitHub](#) • [Email](#) • 03125431964

PROFESSIONAL SUMMARY

Web developer, with two software internships and a portfolio of deployed React, Django and Node.js applications. Final-year project combines a RoBERTa emotion classifier with a locally hosted Llama 3.2 LLM behind a Django REST Framework API, designed so every AI component degrades gracefully rather than failing.

SELECTED PROJECT

AI Cognitive for Mental Health Coaches – Final Year Project

Sept 2025 – June 2026

- Architected and built a full-stack mental health platform using React 19 with TypeScript on the frontend and Django RESTful Framework on the backend, structured across 10 modular apps covering authentication, client and coach profiles, emotion logging, session tracking, AI driven guidance, notifications, appointments, feedback, and resource sharing.
- Built a multi-layer emotion classification system centered on a pretrained RoBERTa model (go emotions, 28 categories), extended with custom logic for detecting sarcasm, negation, idioms, and crisis language. Added a rule-based fallback classifier so the assistant keeps working even if the ML dependencies fail.
- Integrated a locally hosted Ollama LLM (Llama 3.2, 1B and 3B) through a custom prompt-engineering layer that feeds in the detected emotion, intensity, risk level, and the last five conversation turns, so responses stay context-aware and empathetic. Included a template-based fallback for when the LLM is unavailable.
- Designed a rule-based risk assessment engine that scores each interaction based on emotion type, intensity, and crisis-related language, and cross-references it with 90 days of historical trends. Clients who cross a risk threshold automatically trigger a "Professional Support Recommended" alert on the dashboard, prompting human coach intervention.
- Built out the full coach-client workflow, including admin-gated coach license approval, coach assignment, appointment booking with accept, decline, and reschedule options, and a real-time notification system, all supported by a relational schema of 28 tables in Django ORM / SQLite.
- Treated resilience as a core design principle rather than an afterthought. Every AI component (the classifier, the LLM response generator, and the session summarizer) has a built-in fall back, so the system degrades gracefully instead of breaking when something upstream fails.

Tech & AI tools: React • Python (Django) • Ollama Llama 3.2:3b • SQLite3 • RoBERTa • Hugging Face Transformers [\[Github\]](#)

WORK EXPERIENCE

NASTP (National Cybercrime and Forensic Lab)

Intern-Web Development

Air University
July 2025 – Sept 2025

- Developed a comprehensive Browser Forensics Tool using React.js and Python (FASTAPI) to automate the extraction and visualization of critical artifacts, including history, cookies, and login credentials.
- Engineered a real-time monitoring system that tracks live browsing activity, reducing data extraction and logging latency from under 0.1 milliseconds via instant UI notifications.
- Designed a modular backend architecture to parse and process SQLite databases from multiple browser engines, ensuring structured data delivery via RESTful APIs for seamless frontend integration.
- Architected an interactive Timeline Builder within the UI, allowing forensic analysts to chronologically reconstruct user sessions and identify patterns of suspicious activity at a glance.
- [Github](#)

Speridian Technologies

Intern-Web Development

Lahore
July 2024 – September 2024

- Developed a full-stack performance management web application using Angular and TypeScript on the frontend and ASP.NET Core Web API on the backend, creating a responsive and user-friendly experience for employee registration, role management, and performance plan submissions.
- Designed and implemented core modules for employee management, manager assignment, and performance review workflows, improving the usability and structure of the system.
- Participated in the development of a 'Performance Management Plan' system, engineering pipelines to track execution metrics across 50+ active projects and reducing reporting latency by 80%.
- Built and integrated RESTful API services for users, roles, and performance-related data, ensuring smooth communication between the frontend interface and backend operations.
- Managed database design and data operations using SQL Server with Entity Framework Core, supporting efficient storage, retrieval, and maintenance of application data.
- Gained practical experience in web development, debugging, performance tuning, and integrating front-end with backend services, strengthening full-stack development skills.
- [Github](#)

PROJECTS

The Trinket Bloom – Online Shop

- Conceptualized and built a mobile-first e-commerce platform for a handcrafted resin jewellery brand, architecting a decoupled React SPA + custom Express.js RESTful API (SQLite-backed) for real-time data persistence.
- Designed a dynamic product catalogue with search, filtering, and sorting, delivering an intuitive discovery-to-purchase browsing experience.
- Implemented a complete customer journey wishlist, cart, and secure checkout (COD/bank transfer) plus an integrated feedback/contact module.
- Built a modular, responsive UI and deployed both frontend and backend independently on Vercel, ensuring a fast, seamless experience across devices.

Tech: React • NodeJS • SQLite3 [Website](#) [Github](#)

California Boutique – Online Shop

- Designed and delivered a mobile-first e-commerce storefront for Boutique, translating a high-end brand aesthetic into an intuitive, conversion-focused user journey from product discovery to checkout.
- Built the frontend on a modern React + Vite architecture with Tailwind CSS, delivering a responsive, modular UI with fast builds and a maintainable, component-driven codebase.

- Developed a Node.js/Express backend API with dedicated route modules for products, collections, cart, testimonials, and newsletter subscriptions.
- Integrated frontend and backend through a RESTful API layer, delivering dynamic product data and cart functionality across a seamless, high-performance customer experience.

Tech: React • NodeJS [Website](#) [Github](#)

Real Estate

- Built a full-stack real estate web application using React 19 + Vite (frontend) and Node.js + Express 5 (backend) in a pnpm monorepo, with PostgreSQL/Drizzle ORM for data persistence including property listings and a contact-lead capture system.
- Designed a type-safe API contract using OpenAPI as the single source of truth, auto-generating Zod validators for the backend and React Query hooks for the frontend via code generation (Orval) — eliminating drift between client and server.
- Deployed and debugged a production serverless architecture on Vercel, resolving deep TypeScript/module-resolution incompatibilities between a pnpm workspace and Vercel's build system, and provisioning a managed Neon PostgreSQL database — shipping both frontend and backend as independent, live production deployments.

Tech: React • NodeJS [Website](#) [Github](#)

Portfolio

- Architected a fully responsive, component-based portfolio (React 19 + Vite 7 + Tailwind CSS 4) with client-side routing, custom Dark/Light theming, smooth scroll navigation, and a dynamic animated background — built mobile-first for accessibility and cross-device engagement.
- Built an interactive Skills Visualization module and a Featured Projects gallery with tagged tech stacks, live demo links, and direct GitHub repo links, plus a certificates showcase and downloadable resume/CV.
- Ensured code quality and consistency with ESLint (React Hooks + Fast Refresh rules) and composable UI patterns via Radix UI primitives and class-variance-authority/tailwind-merge.
- Deployed and continuously hosted on Vercel with an optimized Vite production build and SEO-friendly meta tags, delivering a fast, polished, professional digital presence.
- [Portfolio](#)

EDUCATION

Foundation University Rawalpindi Campus

Bachelors in Computer Science

Rawalpindi, Pakistan

September 2022 – June 2026

Final-year project: AI Cognitive for Mental Health Coaches.

CORE TECHNICAL SKILLS

Application Development: ReactJS • Angular • TypeScript • dotnet • RESTful APIs

• Python (Django)

AI & Data Science: LLM & Infrastructure: Ollama (Llama 3.2:3B) • NLP Models: RoBERTa • Frameworks & Libraries: Hugging Face Transformers

Methodology & Tools: Agile • ITIL • Git & Github • Postman